

Norway

SLEIPNER

Volve F

F-14

F-14

Plan: F14 Rev6 from 958CP 270508 CC

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Inspirer Rotary Table @ 54.90m (Actual Rotary Table)
Site:	Volve F	MD Reference:	Inspirer Rotary Table @ 54.90m (Actual Rotary Table)
Well:	F-14	North Reference:	Grid
Wellbore:	F-14	Survey Calculation Method:	Minimum Curvature
Design:	F14 Rev6 from 958CP 270508 CC	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude: 58° 26' 29.8068 N
From: Map		Easting:	435,050.02 m	Longitude: 1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence: -0.95 °

Well	F-14			
Well Position	+N/-S	-1.21 m	Northing:	6,478,562.31 m
	+E/-W	2.42 m	Easting:	435,052.44 m
Position Uncertainty	0.00 m	Wellhead Depth:	91.00 m	Water Depth: 91.00 m

Wellbore	F-14				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	MNETICREFERENCE	04.11.2007	-2.23	71.63	50,339

Design	F14 Rev6 from 958CP 270508 CC			
Audit Notes:				
Version:	Phase:	PLAN	Tie On Depth:	2,788.00
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)
	145.90	-1.21	2.42	61.23

Survey Tool Program	Date	19.11.2015		
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description
155.09	2,655.90	F-14 12 1/4" Drop Gyro (F-14)	Wellbore Surveyor, stat	Gyro Tool from GD
2,673.90	2,788.00	F-14 12 1/4" MWD Survey after Drop Gyro	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)
2,788.00	3,779.18	F14 Rev6 from 958CP 270508 CC (F-14)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)

Planned Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
2,788.00	40.19	55.74	2,731.59	152.69	174.94	225.30	0.000	0.000	0.000	
2,798.00	40.19	55.74	2,739.23	156.32	180.28	231.73	0.000	0.000	0.000	
2,965.15	62.48	55.74	2,843.00	229.33	287.47	360.82	4.001	4.001	0.000	
3,253.70	82.10	86.89	2,932.35	311.75	544.12	625.46	3.679	2.040	3.239	
3,276.28	82.10	86.89	2,935.46	312.96	566.45	645.62	0.000	0.000	0.000	
3,326.01	76.00	86.95	2,944.90	315.58	615.19	689.60	3.679	-3.679	0.034	
3,384.97	70.10	86.96	2,962.08	318.57	671.48	740.38	3.004	-3.004	0.007	

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Planned Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
3,423.12	70.10	86.96	2,975.07	320.47	707.30	772.70	0.000	0.000	0.000	
3,475.06	65.00	88.00	2,994.90	322.59	755.24	815.74	2.995	-2.944	0.599	
3,630.09	49.23	95.86	3,078.89	319.02	884.81	927.59	3.304	-3.051	1.522	
3,673.86	49.23	95.86	3,107.48	315.63	917.79	954.87	0.000	0.000	0.000	
3,674.11	49.21	95.88	3,107.64	315.61	917.98	955.03	3.287	-2.952	1.909	
3,685.71	49.22	95.88	3,115.22	314.71	926.72	962.25	0.017	0.016	-0.008	
3,690.36	49.22	95.88	3,118.25	314.35	930.22	965.15	0.000	0.000	0.000	
3,708.18	49.18	95.90	3,129.90	312.97	943.64	976.25	0.068	-0.061	0.039	
3,779.18	49.18	95.90	3,176.31	307.44	997.08	1,020.44	0.000	0.000	0.000	

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Design Targets									
Target Name									
- hit/miss target	Dip Angle	Dip Dir.	TVD	+N/-S	+E/-W	Northing	Easting	Latitude	Longitude
- Shape	(°)	(°)	(m)	(m)	(m)	(m)	(m)		
F14 stying Shallow	0.00	0.00	2,823.57	248.56	347.10	6,478,812.00	435,397.00	58° 26' 38.0241 N	1° 53' 36.0678 E
- plan misses target center by 47.04m at 3019.37m MD (2866.70 TVD, 254.43 N, 329.24 E)									
- Point									
F14 stying: Top 060307	0.00	0.00	2,834.90	256.57	365.11	6,478,820.00	435,415.00	58° 26' 38.2923 N	1° 53' 37.1695 E
- plan misses target center by 44.91m at 3041.89m MD (2875.70 TVD, 263.64 N, 347.71 E)									
- Point									
F14 stying Deep	0.00	0.00	2,840.90	261.57	376.11	6,478,825.00	435,426.00	58° 26' 38.4598 N	1° 53' 37.8426 E
- plan misses target center by 44.03m at 3055.38m MD (2880.84 TVD, 268.80 N, 359.07 E)									
- Point									
F14 T1 Rev6	4.00	300.00	2,842.90	231.86	289.08	6,478,795.30	435,339.00	58° 26' 37.4534 N	1° 53' 32.5088 E
- plan misses target center by 1.81m at 2967.34m MD (2844.01 TVD, 230.43 N, 289.08 E)									
- Rectangle (sides W55.40 H49.22 D0.00)									
F14 RTD Target	0.00	0.00	2,842.90	233.87	324.35	6,478,797.31	435,374.26	58° 26' 37.5372 N	1° 53' 34.6807 E
- plan misses target center by 22.43m at 3000.33m MD (2858.69 TVD, 246.07 N, 314.12 E)									
- Rectangle (sides W38.00 H64.00 D0.00)									
F-14 rev6_target1_Top I	4.00	300.00	2,842.90	231.86	289.08	6,478,795.30	435,339.00	58° 26' 37.4534 N	1° 53' 32.5088 E
- plan misses target center by 1.81m at 2967.34m MD (2844.01 TVD, 230.43 N, 289.08 E)									
- Polygon									
Point 1			2,840.41	-9.04	35.93	6,478,786.31	435,374.84		
Point 2			2,841.91	-36.75	-4.75	6,478,758.58	435,334.22		
Point 3			2,845.39	9.04	-35.93	6,478,804.29	435,303.16		
Point 4			2,843.89	36.75	4.75	6,478,832.02	435,343.78		
F14 RTD Target 250907	0.00	0.00	2,842.90	233.56	323.09	6,478,797.00	435,373.00	58° 26' 37.5265 N	1° 53' 34.6034 E
- plan misses target center by 21.78m at 2999.08m MD (2858.15 TVD, 245.51 N, 313.15 E)									
- Point									
F14 Revised 260307	0.00	0.00	2,845.90	236.56	330.09	6,478,800.00	435,380.00	58° 26' 37.6272 N	1° 53' 35.0319 E
- plan misses target center by 22.77m at 3007.35m MD (2861.68 TVD, 249.21 N, 319.65 E)									
- Point									
Volve P50 Topp	0.00	0.94	2,859.00	96.51	266.07	6,478,660.00	435,316.00	58° 26' 33.0675 N	1° 53' 31.2283 E
- plan misses target center by 116.10m at 2902.56m MD (2810.15 TVD, 199.38 N, 243.49 E)									
- Point									
F14 stying: flt	0.00	0.00	2,924.90	336.59	545.17	6,478,900.00	435,595.00	58° 26' 40.9740 N	1° 53' 48.1864 E
- plan misses target center by 25.93m at 3255.05m MD (2932.54 TVD, 311.82 N, 545.45 E)									
- Point									
F14 Base Hugin Upper I	4.00	300.00	2,944.90	305.58	615.19	6,478,869.00	435,665.00	58° 26' 40.0090 N	1° 53' 52.5337 E
- plan misses target center by 9.98m at 3325.49m MD (2944.77 TVD, 315.55 N, 614.68 E)									
- Rectangle (sides W24.61 H27.70 D0.00)									
F14 Guide Point Fault	0.00	0.00	2,985.22	322.59	755.24	6,478,886.00	435,805.00	58° 26' 40.6327 N	1° 54' 1.1484 E
- plan misses target center by 8.79m at 3471.36m MD (2993.35 TVD, 322.47 N, 751.89 E)									
- Point									
Guidepoint Fault Fallbac	0.00	0.00	2,994.90	322.59	755.24	6,478,886.00	435,805.00	58° 26' 40.6327 N	1° 54' 1.1484 E
- plan hits target center									
- Point									
F14 Stying : TD	0.00	0.00	2,994.90	374.61	683.22	6,478,938.00	435,733.00	58° 26' 42.2755 N	1° 53' 56.6566 E
- plan misses target center by 61.08m at 3409.95m MD (2970.59 TVD, 319.82 N, 694.94 E)									
- Point									
Volve P51 Bunn	0.00	0.94	3,016.02	578.68	423.63	6,479,142.00	435,473.50	58° 26' 48.7323 N	1° 53' 40.4499 E
- plan misses target center by 303.06m at 3248.51m MD (2931.62 TVD, 311.44 N, 538.99 E)									
- Point									
F14 Base Hugin	0.00	0.00	3,081.90	317.09	871.08	6,478,880.50	435,920.80	58° 26' 40.5160 N	1° 54' 8.2937 E

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- plan misses target center by 11.58m at 3620.21m MD (3072.51 TVD, 319.75 N, 877.31 E)									
- Point									
Block 2 Base hugin 1404	0.00	0.00	3,107.90	315.59	918.30	6,478,879.00	435,968.00	58° 26' 40.4925 N	1° 54' 11.2054 E
- plan misses target center by 0.02m at 3674.52m MD (3107.91 TVD, 315.58 N, 918.29 E)									
- Point									
Base Hugin Rev 5 22046	26.00	62.00	3,129.90	312.97	943.64	6,478,876.38	435,993.33	58° 26' 40.4211 N	1° 54' 12.7697 E
- plan hits target center									
- Polygon									
Point 1			3,133.19	-23.95	21.23	6,478,852.08	436,013.88		
Point 2			3,118.71	-19.86	-18.34	6,478,857.74	435,977.28		
Point 3			3,129.30	23.19	-13.89	6,478,899.63	435,979.57		
Point 4			3,143.77	19.10	25.68	6,478,893.97	436,016.17		
Rev-6 Base Hugin	26.00	62.00	3,129.90	312.59	943.31	6,478,876.00	435,993.00	58° 26' 40.4087 N	1° 54' 12.7498 E
- plan misses target center by 0.45m at 3707.96m MD (3129.76 TVD, 312.98 N, 943.47 E)									
- Rectangle (sides W39.78 H43.28 D0.00)									

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
197.70	197.70	30"	30.000	36.000	
1,083.84	1,083.40	20"	20.000	26.000	
2,275.50	2,274.15	14"	14.000	17.500	
2,783.47	2,728.13	9 5/8"	9.625	12.250	

Formations					
Measured Depth (m)	Vertical Depth (m)	Name	Lithology	Dip (°)	Dip Direction (°)
882.34	881.90	Utsira Fm. Top			
1,067.34	1,066.90	Utsira Fm. Base			
1,209.34	1,208.90	Skade Fm. Top			
1,349.34	1,348.90	Skade Fm. Base			
2,063.35	2,062.90	Grid Fm. Top			
2,155.37	2,154.90	Grid Fm. Base			
2,243.93	2,242.90	Balder Fm. Top			
2,312.66	2,310.90	Sele Fm. Top			
2,367.58	2,364.90	Lista Fm. Top			
2,483.82	2,476.90	Ty Fm. Top			
2,608.61	2,589.90	Ekofisk Fm. Top			
2,627.54	2,605.90	Tor Fm. Top			
2,772.70	2,719.90	Hod Fm. Top			
2,863.57	2,785.90	Blodoeks Fm. Top			
2,893.73	2,804.90	Hidra Fm. Top			
2,900.44	2,808.90	Roedby Fm. Top			
2,916.10	2,817.90	Aasgard Fm. Top			
2,944.26	2,832.90	Draupne Fm. Top			
2,960.65	2,840.90	Heather Fm. Top			
2,964.94	2,842.90	Hugin Fm. Top			
3,486.64	2,999.90	Sleipner/Skagerrak Fm. Top			

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Checked By: _____	Approved By: _____	Date: _____
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